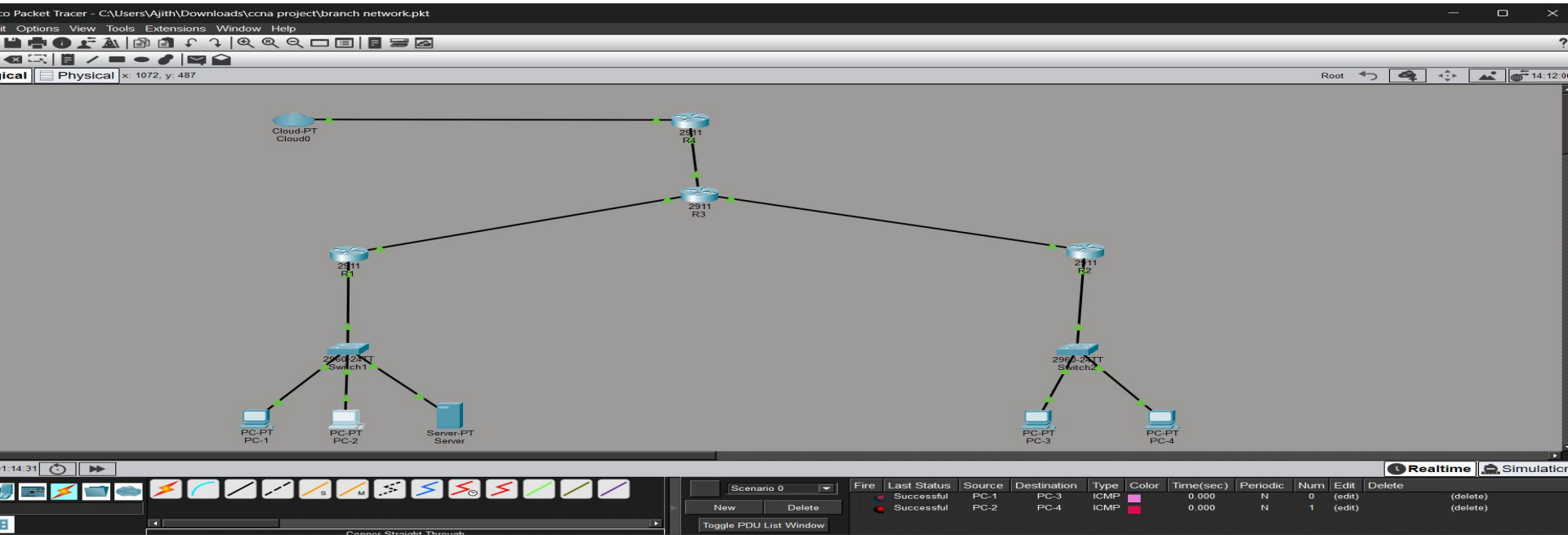


Enterprise Multi-Branch Network

TOPOLOGY

- 4 routers: R1, R2, R3, R4 (all Cisco 2911)
- R3 = core router connecting all 3 domains
- R4 = edge router connected to Cloud-PT (ISP)
- 2 switches + 4 PCs + 1 server at branch LANs



BGP VERIFICATION

Command: show ip bgp summary

- ❖ R3 (AS 65001) ↔ R4 (AS 65000)
- ❖ State: Established | 4 prefixes received

```
R4#
R4#show ip bgp summary
BGP router identifier 4.4.4.4, local AS number 65000
BGP table version is 4, main routing table version 6
3 network entries using 396 bytes of memory
3 path entries using 156 bytes of memory
2/2 BGP path/bestpath attribute entries using 368 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 1000 total bytes of memory
BGP activity 3/0 prefixes, 3/0 paths, scan interval 60 secs

Neighbor      V    AS MsgRcvd MsgSent   TblVer  InQ  OutQ Up/Down  State/PfxRcd
30.0.0.1      4 65001     40      38        4    0    0 00:36:23        4

R4#
```

ROUTE DISTRIBUTIONS

R3's routing table shows:

- ❖ **B** → **BGP** routes from R4
- ❖ **D EX** → **EIGRP** routes redistributed
- ❖ **O E2** → **OSPF** routes redistributed
- ❖ **All 3 domains** see each other's networks

```
R3#
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is 30.0.0.2 to network 0.0.0.0

    3.0.0.0/32 is subnetted, 1 subnets
C       3.3.3.3/32 is directly connected, Loopback0
    4.0.0.0/32 is subnetted, 1 subnets
B       4.4.4.4/32 [20/0] via 30.0.0.2, 00:00:00
    10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/30 is directly connected, GigabitEthernet0/0
L       10.0.0.2/32 is directly connected, GigabitEthernet0/0
    20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/30 is directly connected, GigabitEthernet0/2
L       20.0.0.1/32 is directly connected, GigabitEthernet0/2
    30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       30.0.0.0/30 is directly connected, GigabitEthernet0/1
--More--
```

EIGRP VERIFICATION

Command: show ip eigrp neighbors

- ❖ Result: Neighbor 20.0.0.2, AS 100, Hold 13
- ❖ R2 ↔ R3 adjacency confirmed

```
R3>enable
R3#show ip int br
Interface                IP-Address      OK? Method Status      Protocol
GigabitEthernet0/0       10.0.0.2        YES manual  up          up
GigabitEthernet0/1       30.0.0.1        YES manual  up          up
GigabitEthernet0/2       20.0.0.1        YES manual  up          up
Loopback0                 3.3.3.3         YES manual  up          up
Vlan1                     unassigned      YES unset   administratively down down
R3#
R3#
R3#
R3#
R3#
R3#show ip ospf neighbor

Neighbor ID      Pri   State           Dead Time   Address        Interface
1.1.1.1          1     FULL/BDR        00:00:34    10.0.0.1       GigabitEthernet0/0
R3#
R3#
R3#
R3#show ip eigrp neighbors
IP-EIGRP neighbors for process 100
H   Address          Interface        Hold Uptime      SRTT   RTO   Q    Seq
                               (sec)           (ms)                  Cnt   Num
0   20.0.0.2          Gig0/2           11   00:32:21    40     1000   0     5
```

OSPF VERIFICATION

- ❖ **Command:** show ip ospf neighbor
- ❖ **Result:** Neighbor 3.3.3.3, State FULL/BDR
- ❖ **R1 ↔ R3 adjacency confirmed**

```
R1#  
R1#  
R1#  
R1#SHOW IP OSPF NEIGHBOR
```

Neighbor ID	Pri	State	Dead Time	Address	Interface
3.3.3.3	1	FULL/DR	00:00:39	10.0.0.2	GigabitEthernet0/0

```
R1#
```

END TO END TEST

- ❖ Ping from branch PC to 4.4.4.4 (R4 loopback)
- ❖ Result: 4 sent, 4 received, 0% loss
- ❖ Confirms full cross-domain connectivity

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Request timed out.
Reply from 192.168.20.10: bytes=32 time=1ms TTL=125
Reply from 192.168.20.10: bytes=32 time=10ms TTL=125|
Reply from 192.168.20.10: bytes=32 time<1ms TTL=125

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 3, Lost = 1 (25% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 10ms, Average = 3ms

C:\>ping 4.4.4.4

Pinging 4.4.4.4 with 32 bytes of data:

Reply from 4.4.4.4: bytes=32 time<1ms TTL=253
Reply from 4.4.4.4: bytes=32 time<1ms TTL=253
Reply from 4.4.4.4: bytes=32 time=2ms TTL=253
Reply from 4.4.4.4: bytes=32 time<1ms TTL=253

Ping statistics for 4.4.4.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 2ms, Average = 0ms

C:\>
```

INTERFACES R1,R2,R3.R4

R1>en

R1#sh ip int br

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	10.0.0.1	YES	manual	up	up
GigabitEthernet0/1	unassigned	YES	manual	administratively down	down
GigabitEthernet0/2	192.168.10.1	YES	manual	up	up
Loopback0	1.1.1.1	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

R1#

R2#sh ip int br

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	20.0.0.2	YES	manual	up	up
GigabitEthernet0/1	192.168.20.1	YES	manual	up	up
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
Loopback0	2.2.2.2	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

R2#

R3>en

R3#sh ip int br

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	10.0.0.2	YES	manual	up	up
GigabitEthernet0/1	30.0.0.1	YES	manual	up	up
GigabitEthernet0/2	20.0.0.1	YES	manual	up	up
Loopback0	3.3.3.3	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

R3#

R4#show ip int br

Interface	IP-Address	OK?	Method	Status	Protocol
GigabitEthernet0/0	30.0.0.2	YES	manual	up	up
GigabitEthernet0/1	8.8.8.1	YES	manual	up	up
GigabitEthernet0/2	unassigned	YES	unset	administratively down	down
Loopback0	4.4.4.4	YES	manual	up	up
Vlan1	unassigned	YES	unset	administratively down	down

THANK YOU